

Raphaël Tuor

PhD in Computer Science | Early
Career Researcher
Research Systems, Prototyping &
Evaluation

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PhD researcher in Computer Science with experience in designing, implementing, and evaluating experimental research systems and prototypes. Background in interactive systems, data visualization, and applied software development, with a strong peer-reviewed publication record.

CORE SKILLS

- Experimental system design and research prototyping
- Software development for research and experimentation
- Programming: JavaScript, C#, Python (basic), Java
- Data processing and visualization
- System evaluation and experimental analysis
- Scientific writing and peer review

PROFESSIONAL EXPERIENCE

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| 2023 – 2025 | Senior Researcher (100%), Human-IST Institute, University of Fribourg
SNSF Project – ALGEPI <ul style="list-style-type: none">• Design and implementation of experimental research prototypes• Development of research software for recommender system experimentation• System evaluation and data-driven analysis• Scientific writing and peer-review activities• Collaboration within interdisciplinary research teams Project website: www.algepi.com
Supervisor: Prof. Denis Lalanne |
| 2018 – 2022 | Research Assistant / Junior Researcher (100%) – Human-IST Institute, University of Fribourg <ul style="list-style-type: none">• Research activities in human-computer interaction• Participation in experimental studies and research projects Supervisor: Prof. Denis Lalanne |
| 2018 & 2020 – 2021 | Civil Service – Association St-Camille, Villars-sur-Glâne <ul style="list-style-type: none">• Assistant Instructor• Development of interactive experimental systems using Unity (C#)• Prototypes presented in public exhibitions and festivals |
| 2020 | VidSummit 2020 – Speaker (online event) |

Public talk: *“Stop being a perfectionist: how to go from 0 to 100k subscribers and make money”*.

- 2015 – 2016 IT Department Member – Junior Entreprise Fribourg**
- Management and development of web projects for various clients
- 2014 Civil Service – Résidence bourgeoise des Bonnesfontaines (Fribourg)**
- IT and administrative support
 - Basic accounting assistance
 - Creation of a musical jingle for the residence
- 2013 – 2014 Research Collaboration – University of Fribourg**
SNSF Project – Veni, Vidi, Ludique
- Creation of audio jingles for QR-code based installations
 - Music production for exhibition films at the Roman Museum of Nyon
 - Collaboration with interdisciplinary research and cultural teams
- Project website: www.venividiludique.ch
- 2012 Web Development Project – Vocal Ensemble “La Villanelle”**
- Website development (HTML, CSS, jQuery, JavaScript)
 - Visual design and integration (Photoshop)
- 2011 Civil Service – Cantonal and University Library of Fribourg**
- Assistant librarian
 - RFID tagging of documents for automated lending systems

EDUCATION

- 2017 – 2023 PhD in Computer Science**, Faculty of Science, University of Fribourg
Dissertation: “Visual Analytics in High-Stakes Decision Making: Insights from Design and Time-Critical Domains”
Supervisor: Prof. Denis Lalanne
- 2014 – 2016 Swiss Joint Master of Science in Computer Science**, Faculty of Science, University of Fribourg
Master’s Thesis: “Enhancing the Visualization of Mixed Multidimensional Data in Parallel Coordinates”
Supervisors: Prof. Denis Lalanne, Florian Evéquoz
Grade: 6 (completed December 2016)
- 2010 – 2014 Bachelor of Science in Computer Science**, Faculty of Science, University of Fribourg
Bachelor’s Thesis: “Data Processing and Visualization of Ecological Graphs”
Supervisors: Nayla Sokhn, Prof. Jean Hennebert, Prof. Denis Lalanne
Grade: 5.5 (completed November 2014)
- 2012 – 2013 Minor in Management Information Systems**, University of Fribourg
Completed during Spring semester 2013
- 2010 – 2013 Minor in Journalism and Media Studies**, University of Fribourg
Completed during Fall semester 2013
- 2006 – 2010 Swiss Matura (High School Diploma)**
Collège St-Michel, Fribourg
Focus: Latin, Philosophy
Final Project: “Biomechanics of a Sports Gesture: Rowing”
Grade: Good

SELECTED PUBLICATIONS

2024

- Fuchs, N., Vanhulst, P., Tuor, R., & Lalanne, D. (2024). **Flowstrates++: An Approach to Visualize Multi-Dimensional OD Data**. In *VISIGRAPP (1): GRAPP, HUCAPP, IVAPP* (pp. 625-636).

2022

- Tuor, R., & Lalanne, D. (2022). **Scenario-Based Simulation Platform for Developing New Interfaces and Benchmarking Controller Performance**. *AIAA SCITECH 2022 Forum*.

2021

- Vanhulst, P., Tuor, R., Évéquoz, F., & Lalanne, D. (2021). **Colvis - A Structured Annotation Acquisition System for Data Visualization**. *Information*, 12(4), 158.

2019

- Vanhulst, P., Évéquoz, F., Tuor, R., & Lalanne, D. (2019). **A descriptive attribute-based framework for annotations in data visualization**. *Communications in Computer and Information Science*, 997, 143-166.
- Jusselme, T., Hoxha, E., Cozza, S., Tuor, R., Züllli, R., Henchoz, N., & Lalanne, D. (2019). **A data-driven approach for lifecycle performance**. Park Books.

2018

- Vanhulst, P., Évéquoz, F., Tuor, R., & Lalanne, D. (2018). **Designing a classification for user-authored annotations in data visualization**. In *VISIGRAPP 2018 - Proceedings of the 13th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications* (Vol. 3, pp. 85-96).
- Tuor, R., Évéquoz, F., & Lalanne, D. (2018). **Parallel bubbles - Evaluation of three techniques for representing mixed categorical and continuous data in parallel coordinates**. In *VISIGRAPP 2018 - Proceedings of the 13th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications* (Vol. 3, pp. 252-263).
- Cozza, S., Jusselme, T., Andersen, M., Tuor, R., & Lalanne, D. (2018). **Usability assessment of building performance simulation tools: A pilot study**. In *Second International Conference for Sustainable Design of the Built Environment - SDBE 2018*.

LANGUAGES

French:	Native	German: Proficient
English:	Fluent	

RESEARCH SOFTWARE DEVELOPMENT & PROTOTYPING

- **Mobile development:** React Native, Xamarin
 - **Interactive systems and game prototyping:** Unity (C#)
 - **Web development:** JavaScript, HTML, CSS, PHP
 - **Data visualization and prototyping:** d3.js, JavaScript, LaTeX
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